

Capstone Project

Project Title:

HR Management Automation Testing Using Playwright (OrangeHRM Demo)

Project Overview:

This project focuses on automating testing of an HR Management System using Playwright. It validates key enterprise workflows such as authentication, employee management, admin operations, leave handling, time tracking, and user profile management.

Website:

OrangeHRM Demo - <https://opensource-demo.orangehrmlive.com/>

Services Covered:

Authentication Service:

- Valid login with credentials
- Invalid login scenarios
- Empty field validation
- Logout functionality
- Session validation after logout

Admin Service:

- Add new users
- Search users
- Edit user roles
- Delete users

- Validate user permissions

PIM (Employee Management):

- Add employee
- Edit employee details
- Search employee
- Delete employee
- Validate employee records

Leave Management:

- Apply leave
- Approve leave
- Reject leave
- Leave balance validation
- Leave history check

Time Module:

- Timesheet submission
- View timesheet
- Edit timesheet
- Validate time entries

Dashboard Service:

- Widget visibility
- Role-based dashboard checks

Profile Service:

- View profile
- Update personal details
- Change password

Logout Service:

- Successful logout
- Session destruction validation
- Back button security check

Tools & Technologies:

Category	Technology
Automation Tool	Playwright
Language	JavaScript
Runtime	Node.js
IDE	VS Code

Category	Technology
Version Control	Git & GitHub

RTM:

Requirement ID	Module	Test Scenario	Test Case Count
REQ-1	Authentication	Login/Logout	12
REQ-2	Admin	User Management	20
REQ-3	PIM	Employee Management	20
REQ-4	Leave	Leave Operations	15
REQ-5	Time	Timesheet	10
REQ-6	Profile	User Profile	10
REQ-7	Dashboard	UI Validation	10
REQ-8	Security	Session Handling	10
REQ-9	Negative Testing	Error Handling	10

Total Test Cases: 120+

Conclusion:

This project provides enterprise-level automation testing using Playwright ensuring stable and scalable QA coverage.